

Photocatalytic Degradation of Methylene Blue Dye by UV Light and TiO₂ Catalyst

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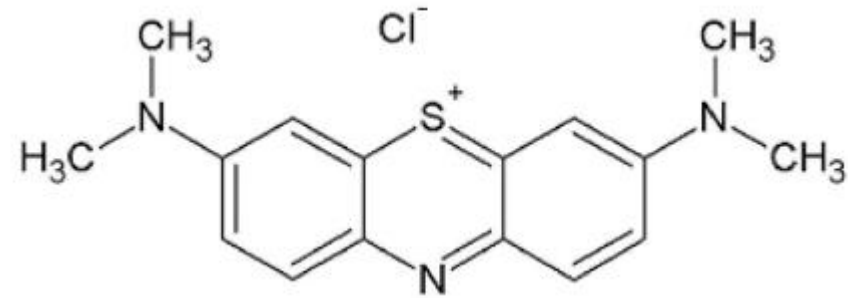
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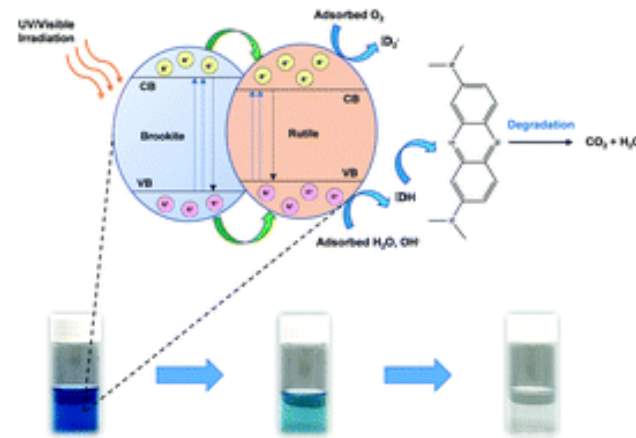
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This presentation focuses on the importance of the breaking down Methylene Blue, the experiment, and the rate law



Why Methylene blue needs to be degraded



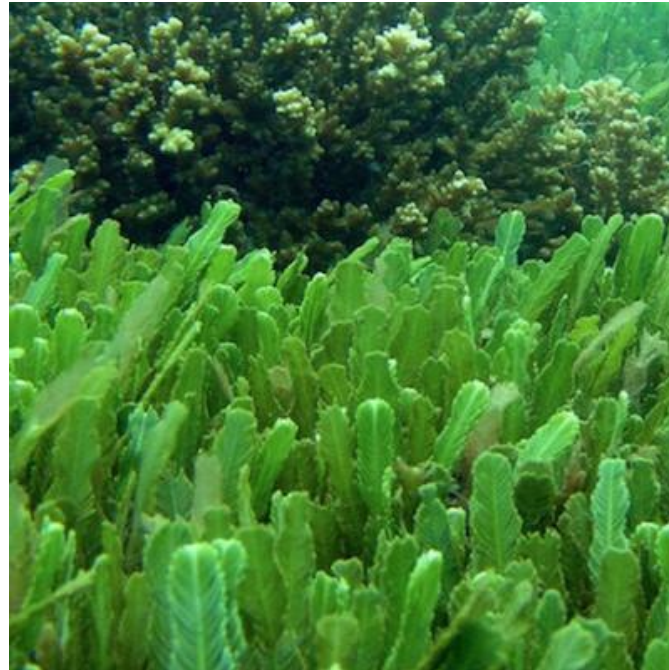
How the experiment was conducted

$$\frac{Ca}{Ca0} = e^{-kt}$$

The observed rate law and the effects of adding hydrogen peroxide

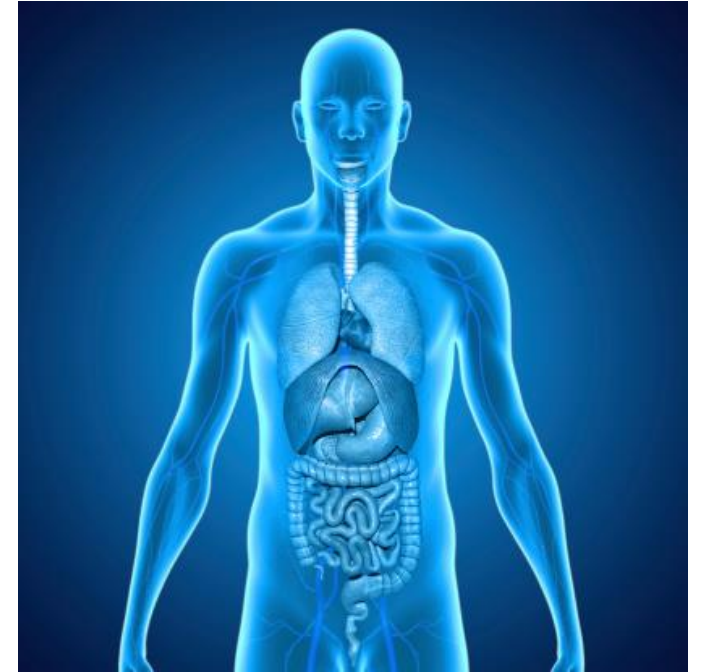
Methylene Blue is toxic in the environment for humans, animals, and the ecosystem

Growth inhibition and reduction of pigment and protein content in types of algae



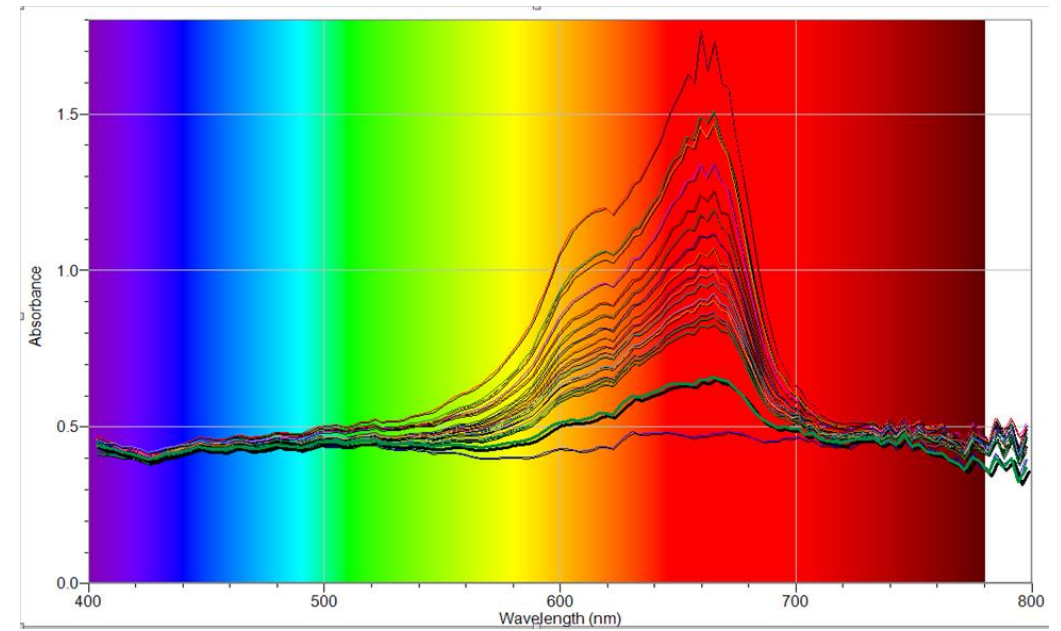
(leshuillettes.com)

Various ailments including tissue necrosis and gangrene

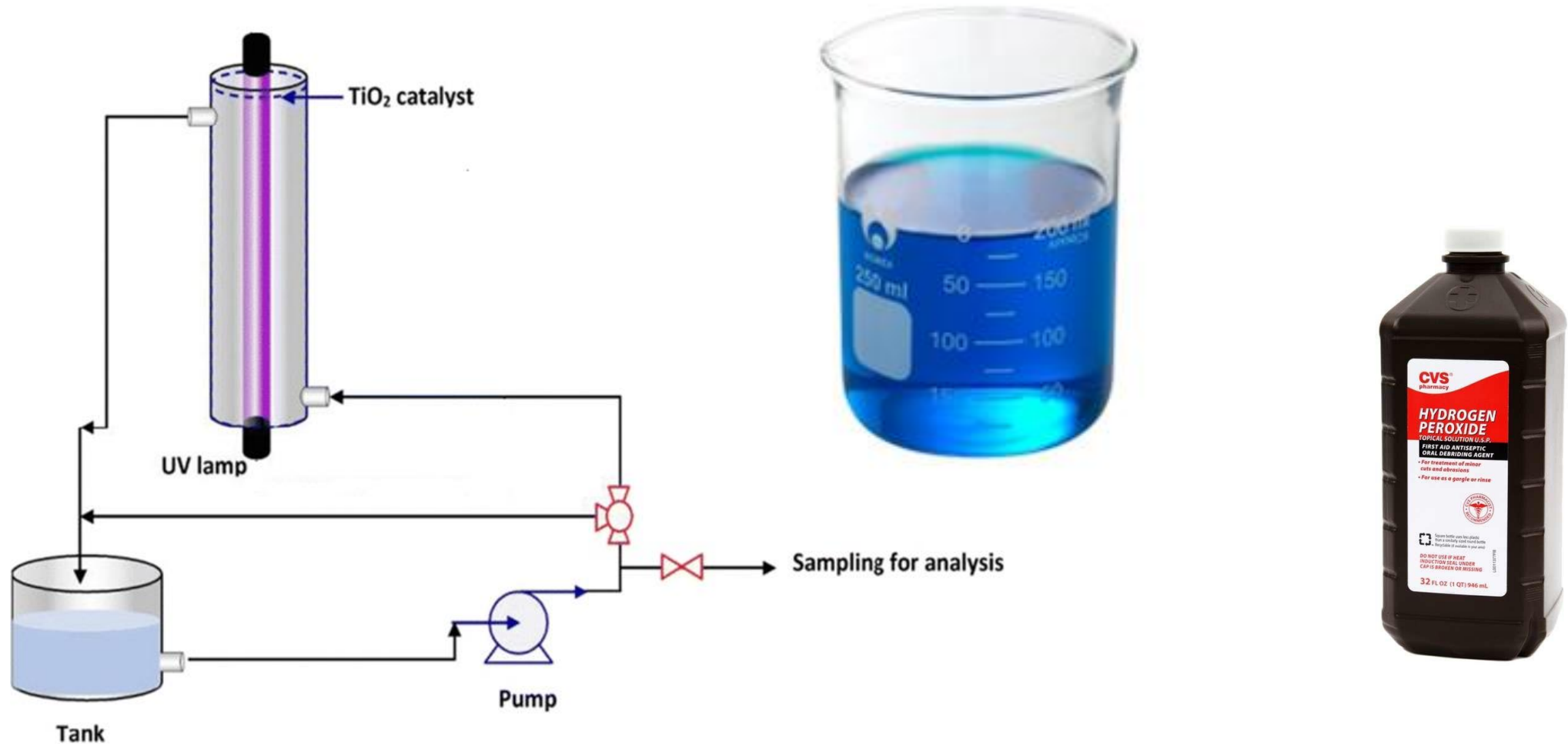


(istockphoto.com)

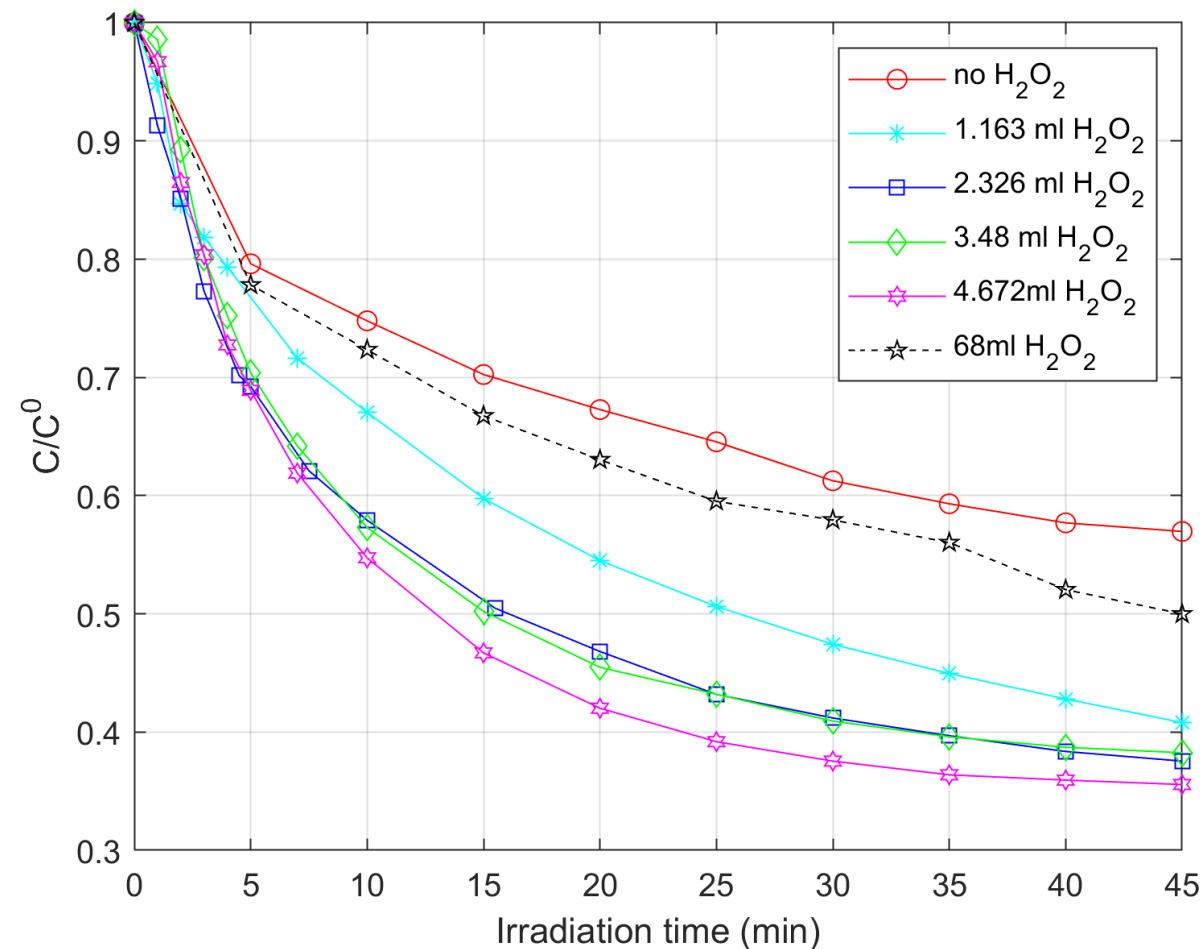
A calibration curve was made to relate the light absorbed to the concentration of the solution



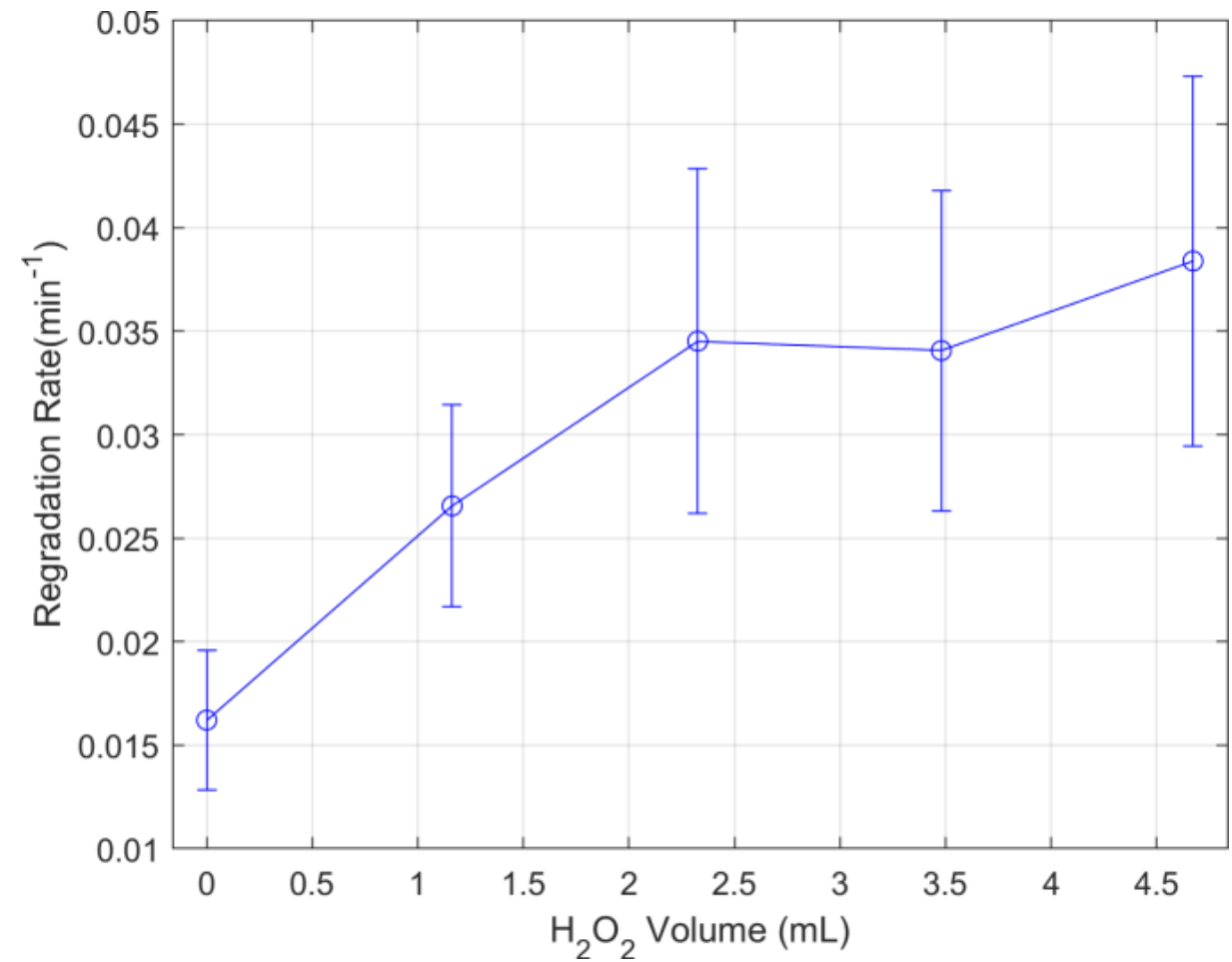
The experiment was conducted using a solution with a starting concentration of 8 ppm methyl blue and varying amounts of hydrogen peroxide.



Adding hydrogen peroxide increases the rate of the reaction

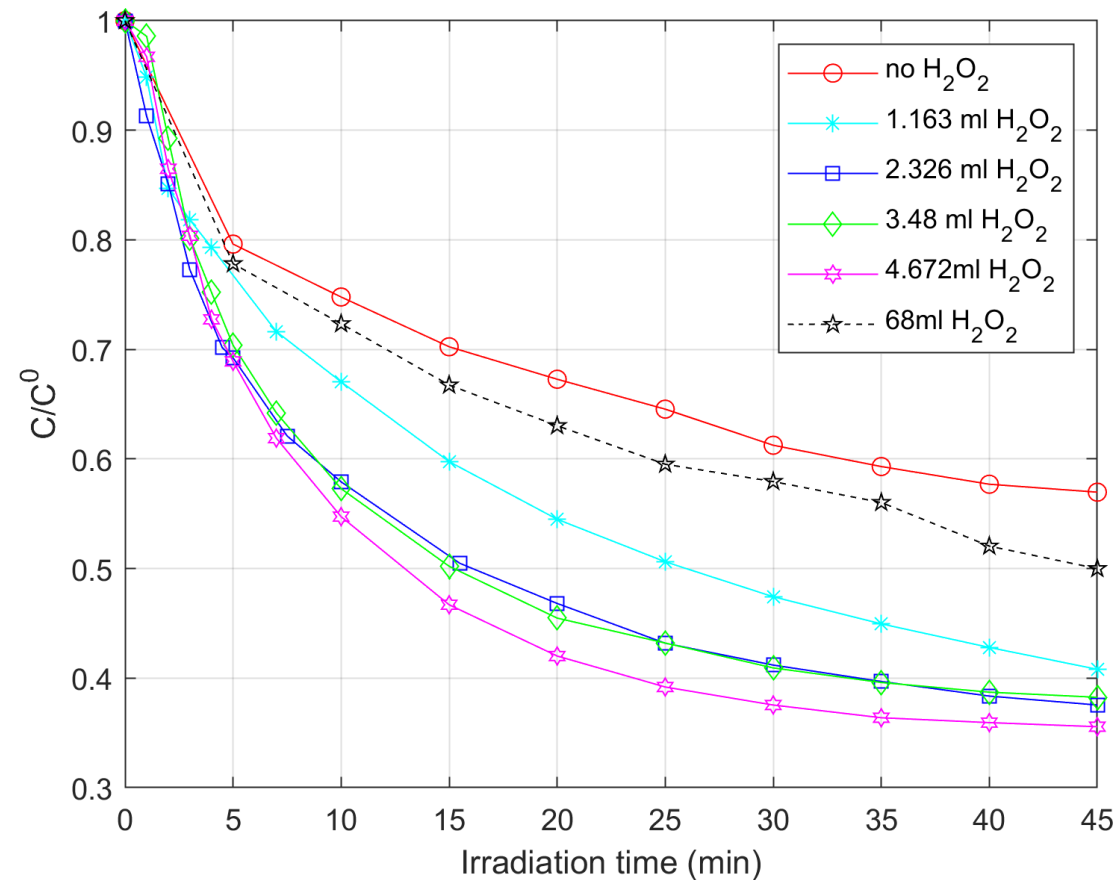


For this experiment 4.6ml of hydrogen peroxide was optimal



In summary, adding small amount of hydrogen peroxide is an effective way of increasing the degradation reaction

This should be researched further to find the optimal amount of hydrogen peroxide



Questions?